

Dual Motor Control for Robots

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Presented here is a simple circuit that can drive two motors for a small robot, allowing the robot to negotiate an obstacle course. Two light-dependent resistors (LDRs) are used to detect the obstacle and the motors are driven correspondingly to avoid the obstacles automatically. Two H-bridge motor circuits are used that can drive each motor forward or backward, or stop it, independently.



Circuit and working

Fig. 1 shows the circuit of dual motor control. The circuit is built around four-channel multiplexer CD4052 (IC1), light-dependent resistors (LDR1 and LDR2), four BC547 npn transistors (T1 through T4), four BC338 transistors

Test Points

Test point	Details
TP0	0V, GND
TP1	9V
TP2	High (when obstacle detected by LDR1)
TP3	High (when obstacle detected by LDR2)
TP4, TP7	High (when TP2 = TP3 = Low)
TP5	High (when TP2 = High and TP3 = Low)
TP8	High (when TP2 = Low and TP3 = High)
TP6, TP9	High (when TP2 = TP3 = High)

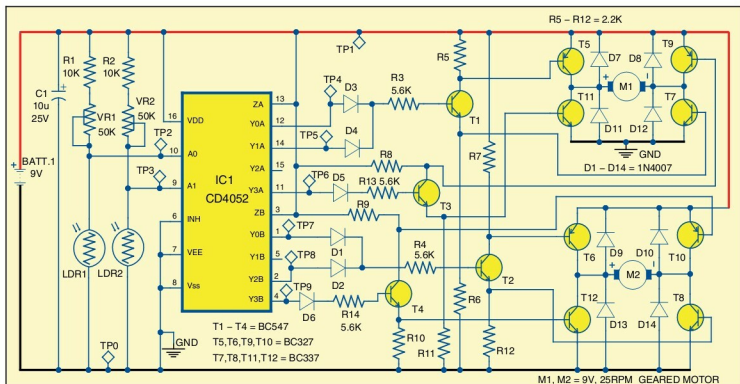


Fig. 1: Circuit of the dual motor control

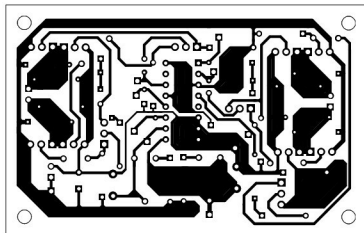


Fig. 2: An actual size, single-side PCB for the dual motor control

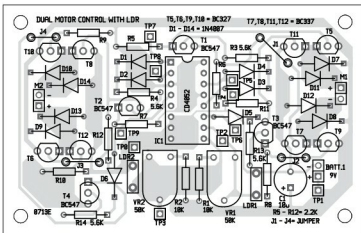


Fig. 3: Component layout for the PCB